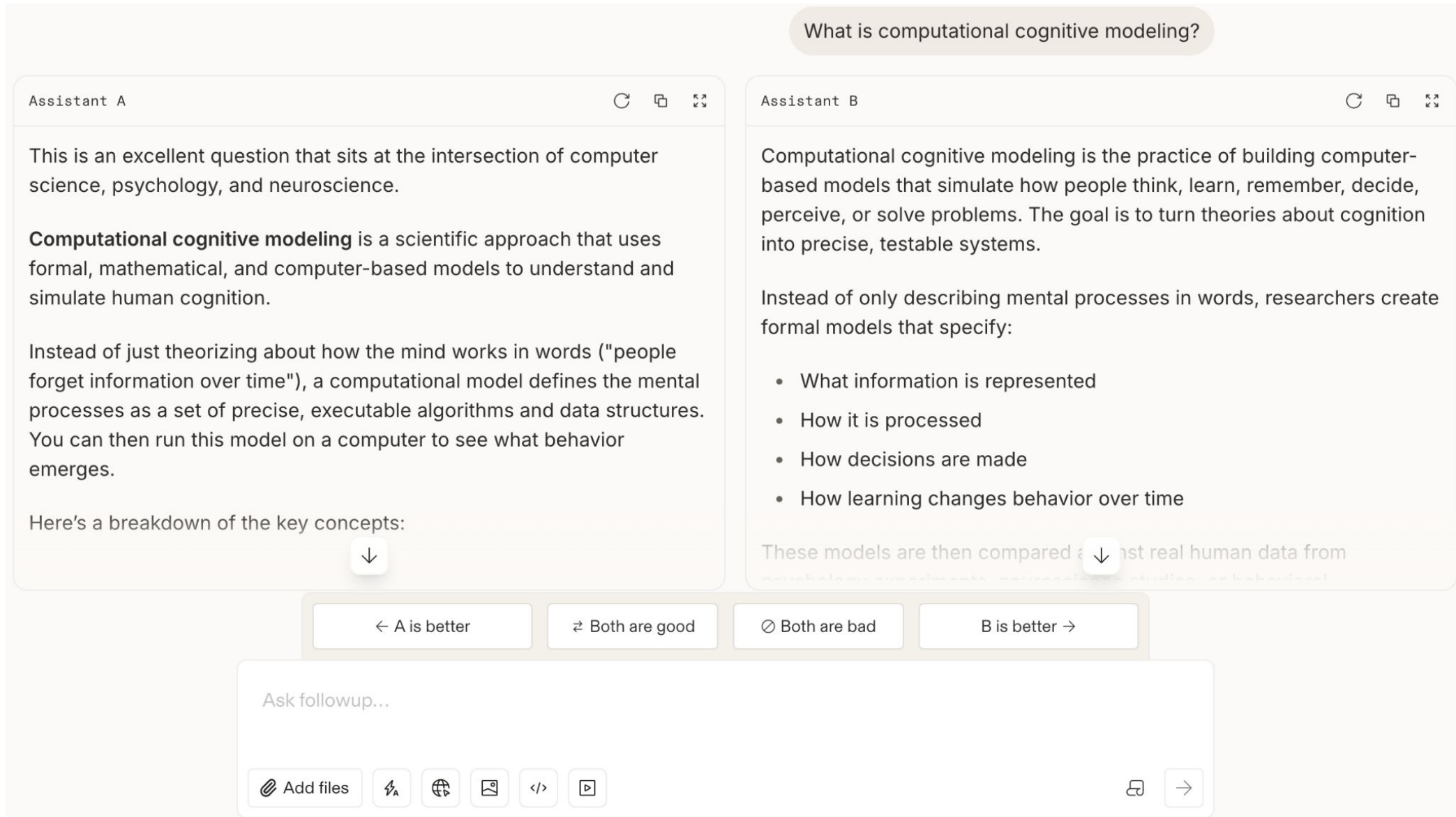


Cognitive Bias in Arena-Style LLM Evaluation: Modeling Human Preferences Beyond Raw Vote

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Introduction



Chatbot Arena is a crowdsourced evaluation platform where users compare two anonymous LLM responses to the same prompt and vote for the better one. Votes are aggregated using a Bradley-Terry model to produce a leaderboard of over 100 models.

Motivation

Chatbot Arena treats **human preference votes** as a **direct signal of model quality**. Yet cognitive science tells us that "preferences are not simply read off some master list... they are constructed" ^[1] and shaped by **context**, **presentation**, and **surface-level cues** rather than content alone.

Goal:

Test whether **observable response features**, such as *length*, *markdown formatting*, and *display position*, systematically shift human preference votes beyond what model identity alone can explain.

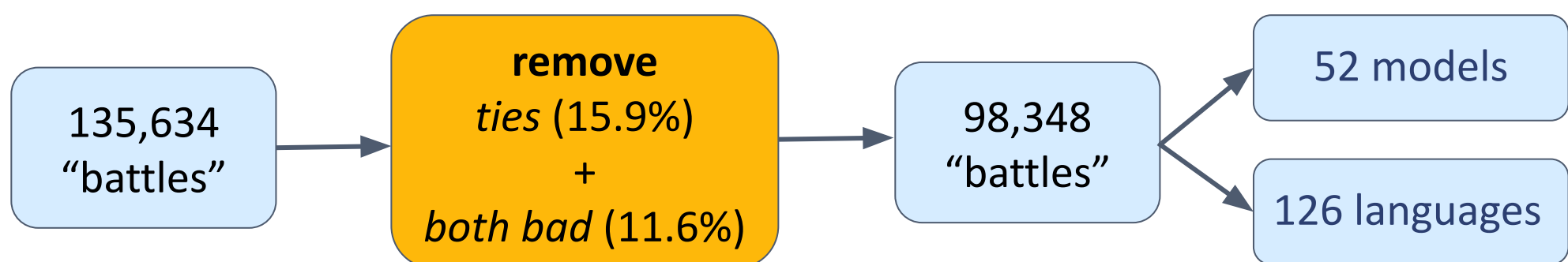
Contributions:

We fit a Bradley-Terry model that separates latent model skill from surface-cue effects, quantify each bias term with uncertainty estimates, and show that controlling for these cues changes individual model ranks by up to 11 positions on the leaderboard.

Dataset



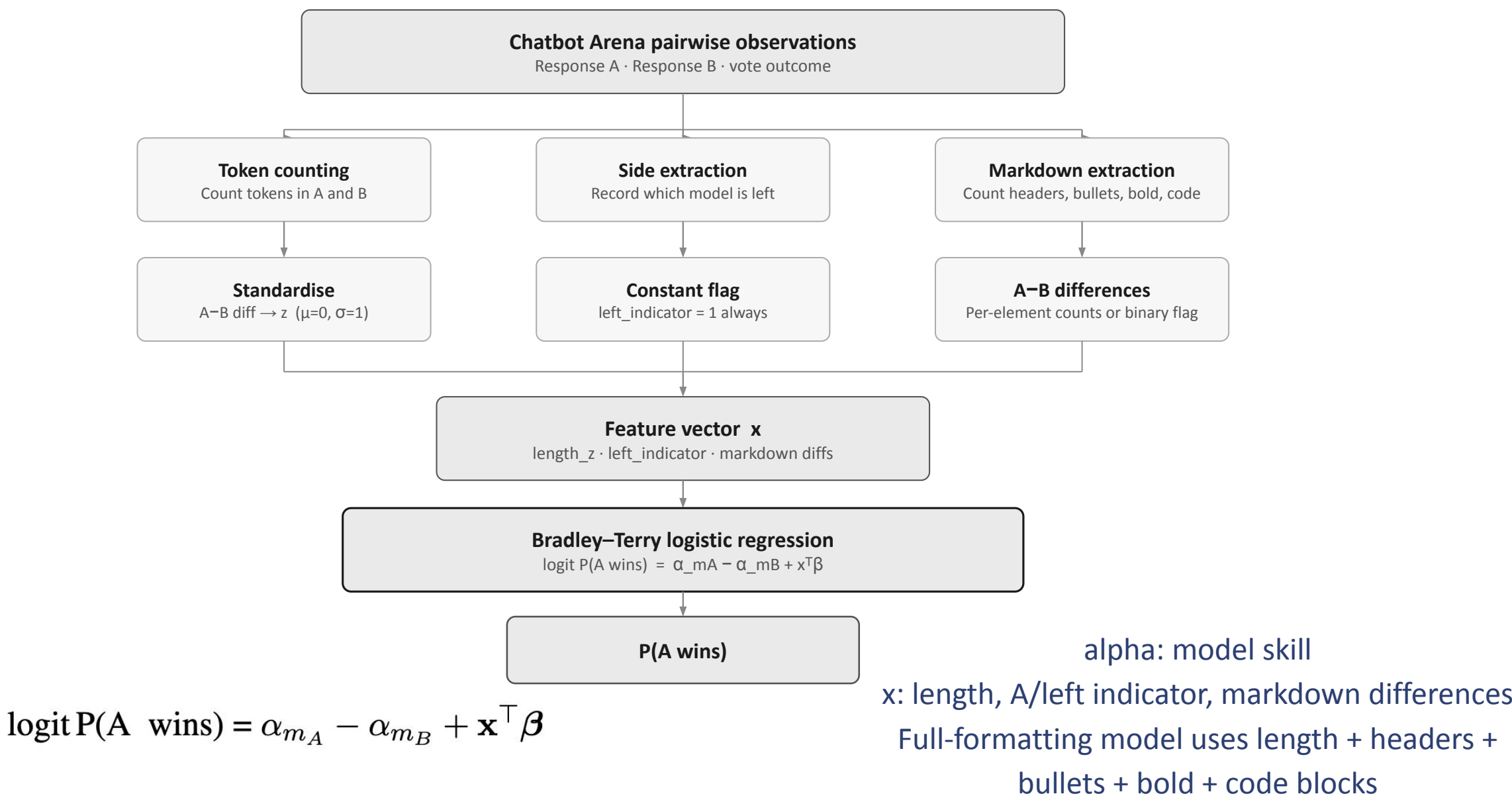
Hugging Face: [Imarena-ai/arena-human-preference-140k](https://huggingface.co/datasets/Imarena-AI/arena-human-preference-140k)



Reference

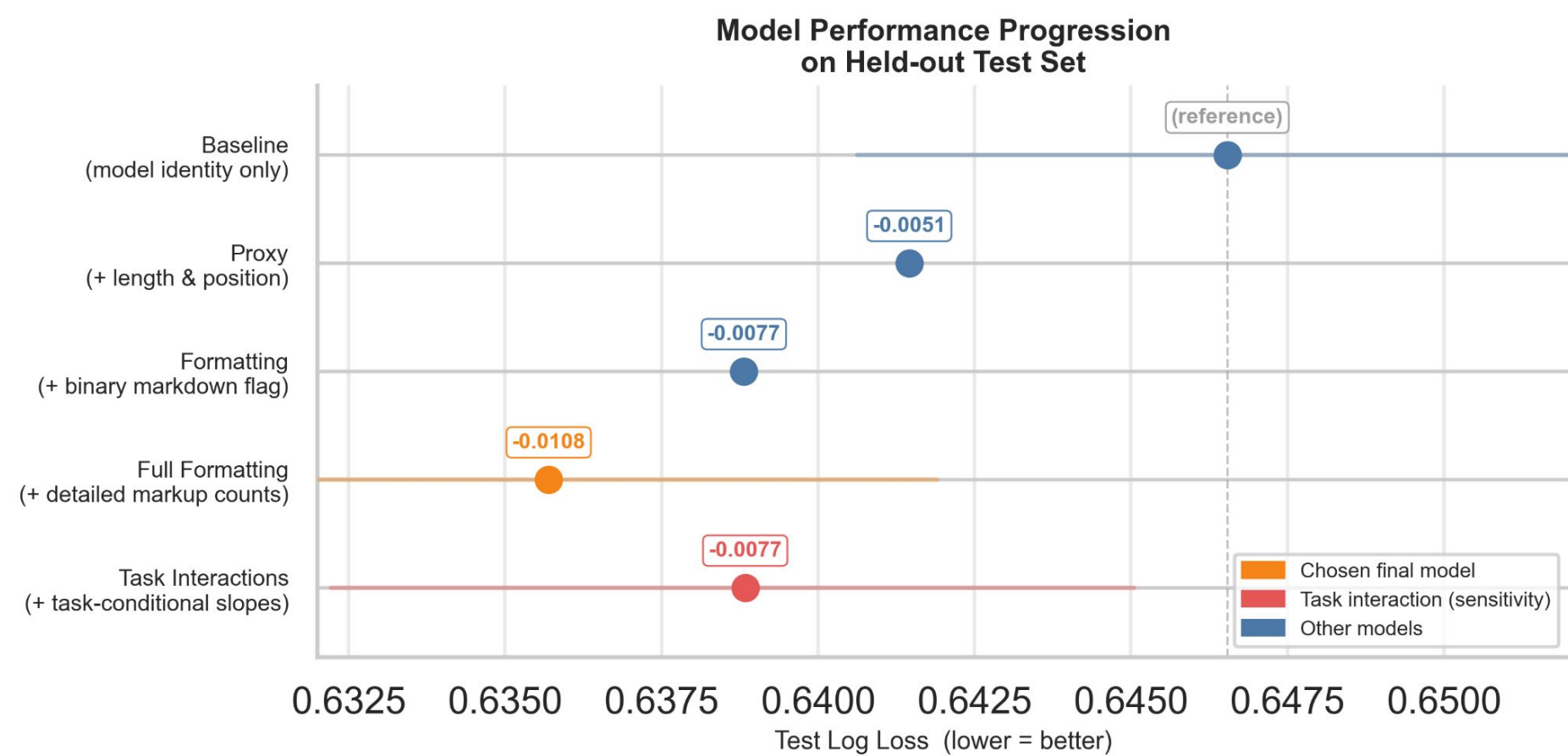
[1] Paul Slovic. 1995. The construction of preference. American Psychologist, 50(5):364–371. <https://doi.org/10.1037/0003-066X.50.5.364>.

Experimental Design



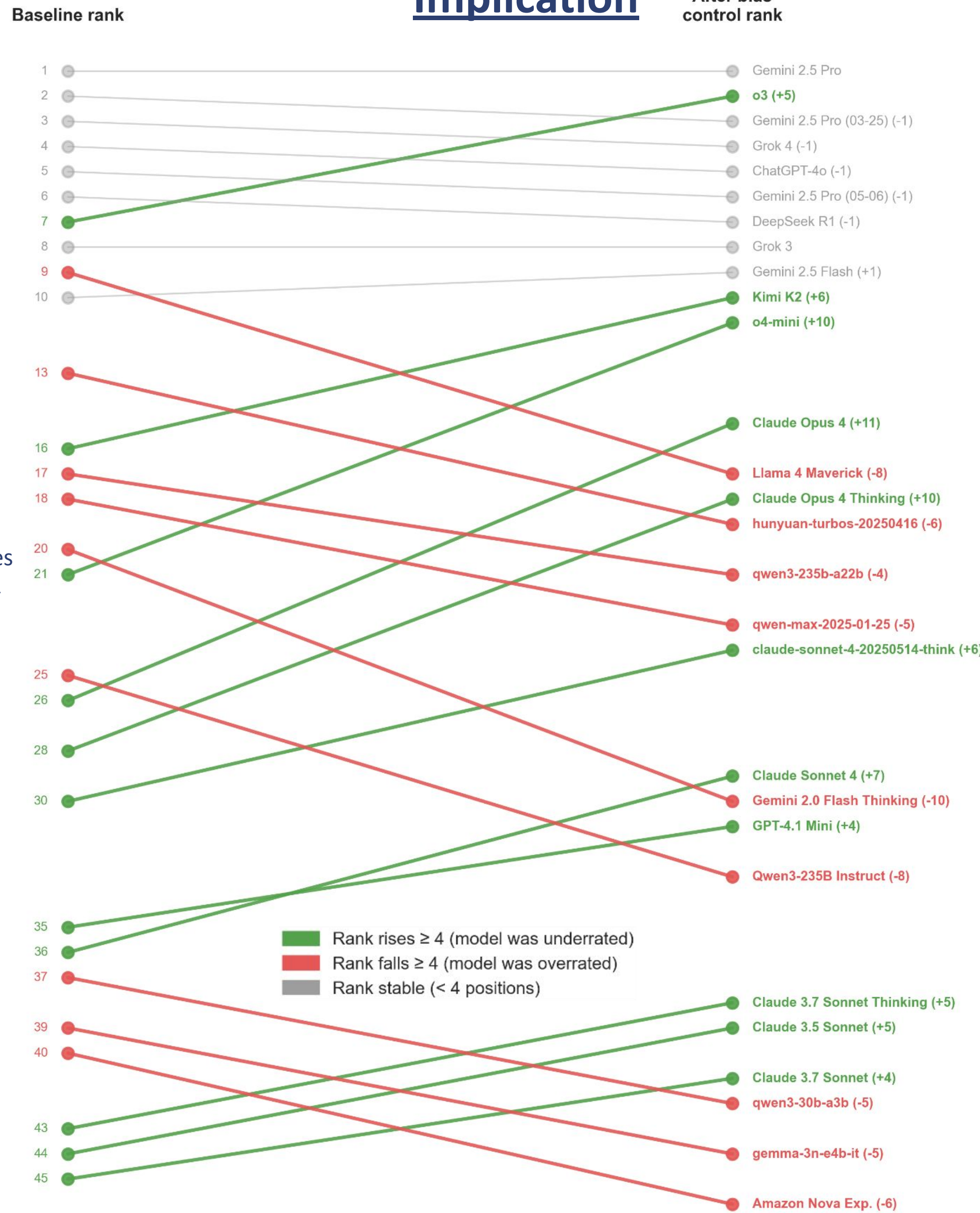
Primary Results

Model	Features added	Log Loss ↓	Accuracy ↑	Brier ↓
Baseline	-	0.6465	0.627	0.2274
Proxy augmented	Length + side indicator	0.6415	0.636	0.2245
Formatting aug.	+ Any-markdown contrast	0.6388	0.641	0.2232
Full formatting	+ 4 markdown count differences	0.6357	0.638	0.2224
Task interaction	Length × markdown × task type	0.6388	0.642	0.2232



Feature	Estimate	Cognitive proxy
Side A indicator	−0.027	Spatial position
Length difference (z)	+0.160	Verbosity heuristic
Header count diff	+0.011	Perceived structure
Bullet count diff	+0.001	Perceived structure
Bold span count diff	+0.010	Perceived structure
Code block count diff	+0.035	Perceived structure

Implication



Controlling for observable surface cues shifts individual model ranks by up to 11 positions. Models that fall in the adjusted ranking (e.g., gemini-2.0-flash-thinking, llama-4-maverick) disproportionately benefit from verbosity and formatting in the raw leaderboard. Models that rise (e.g., claude-opus-4, o4-mini) have skill estimates less confounded by response style. In competitive evaluation contexts where adjacent ranks drive adoption decisions, these differences are practically significant.

Conclusion

Human preference votes in Chatbot Arena conflate model quality with response style. A bias-aware Bradley-Terry model improves held-out calibration over baseline ($\Delta \text{log loss} = -0.0108$), and controlling for length and markdown formatting shifts individual ranks by up to 11 positions. Verbosity ($\beta = 0.160$) and visual structure (code blocks $\beta = 0.035$) act as heuristic proxies for quality, while side position is negligible ($\beta = -0.027$). This aligns with Slovic's (1995) preference construction account preferences are built in the moment from available surface signals. Arena leaderboards measure a mixture of quality and style; surface-adjusted scores offer a complementary, more orthogonal view of true model performance. Future work should test whether LLM-as-judge replicates these same biases, and whether effects differ across languages.